

GraphStream-IDS

Continual Graph Neural Network for Zero-Day Intrusion Detection Under Network Concept Drift

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Abstract

Network intrusion detection systems built on machine learning typically assume that traffic patterns and attack behaviors remain stationary after training, an assumption that fails in real deployments where network conditions evolve and adversaries continuously develop new attack techniques. This phenomenon, known as concept drift, causes static models to degrade over time, while their inability to represent relationships between hosts, ports, and communication sessions further limits their capacity to recognize structurally unusual behavior. This project proposes GraphStream-IDS, a continual-learning intrusion detection framework that represents network traffic as a dynamic graph and combines four complementary mechanisms to address these limitations jointly: a graph neural network encoder (GraphSAGE/GAT) that captures relational structure between network entities, an explicit drift detector that monitors shifts in feature and embedding distributions over time, a selective replay memory that retains representative samples of prior attack classes to mitigate catastrophic forgetting without storing the full historical dataset, and an uncertainty-based novelty detector that flags previously unseen attack behavior rather than forcing it into an incorrect known category. The system is evaluated using a chronological, non-i.i.d. protocol on public benchmark datasets (CIC-IDS2017, CSE-CIC-IDS2018, UNSW-NB15, and TON-IoT), with at least one attack family withheld to simulate a genuine zero-day condition, and is compared against classical tabular classifiers, sequential models, static graph baselines, and continual-learning variants using ordinary replay. Performance is assessed through macro-F1, false-positive rate, zero-day detection rate, AUROC/AUPRC, catastrophic-forgetting score, and memory and latency overhead, supported by ablation studies isolating the contribution of each architectural component. While prior work has separately explored graph-based intrusion detection, drift-aware replay for tabular models, and open-world zero-day detection, no existing framework unifies dynamic graph representation with drift-triggered selective replay and uncertainty-calibrated unknown-attack detection under a single, chronologically evaluated system. GraphStream-IDS is intended as a defensive research contribution, developed and evaluated exclusively on public datasets and controlled simulations, and is targeted for submission as a peer-reviewed research publication.

Keywords

Graph Neural Networks, Continual Learning, Concept Drift, Zero-Day Attack Detection, Intrusion Detection Systems, Selective Replay, Uncertainty Estimation, Network Security